

Customer Shopping Behaviour Analysis

Uncovering insights from 3,900 transactions across product categories — spending patterns, customer segments, and subscription behaviour to guide strategic decisions.

DATA ANALYTICS PROJECT

PYTHON · SQL · POWER BI



Dataset at a Glance

3,900

Rows

Transaction records

18

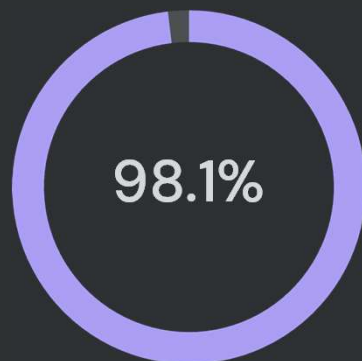
Columns

Dataset fields

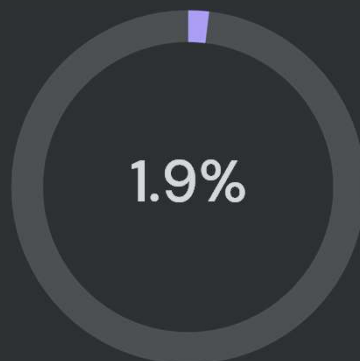
37

Missing Ratings

Review Rating values



Completeness



Missing

Source & Scale

Kaggle retail transactions dataset with 3,900 rows and 18 columns. Each row represents a customer purchase, giving a transaction-level view of shopping behaviour across categories, demographics, and promotions.

What the fields cover

- Demographics: Age, Gender, Location, and Subscription Status
- Purchase details: Item, Category, Amount, Season, Size, and Colour
- Behaviour: Discount, Promo Code, Previous Purchases, Frequency, Review Rating, and Shipping Type

Data quality notes

- 37 missing values in Review Rating
- Mostly structured categorical fields, suitable for segmentation and comparison
- Useful for identifying category mix, discount sensitivity, and repeat-purchase patterns

Category meaning

Clothing and Accessories capture core fashion purchases, while Footwear and Outerwear represent complementary and seasonal categories that can shift with weather and demand cycles.

Data Preparation Process

01

Data Loading

Imported dataset using `pandas`.

02

Initial Exploration

Checked structure and summary statistics.

03

Missing Data Handling

Step 03: addressed missing values in the dataset.

04

Feature Engineering

Created age groups and purchase frequency.

05

Database Integration

Step 05: prepared the data for database integration.

Exploratory Data Analysis in Python

Data Loading & Exploration

Imported via `pandas`; used `df.info()` and `.describe()` for structure and summary statistics.

Missing Data Handling

Imputed 37 missing Review Rating values using the **median rating per product category**.

Feature Engineering

Created `age_group` (binned ages) and `purchase_frequency_days`; dropped redundant `promo_code_used`.

	customer_id	age	gender	item_purchased	category	purchase_amount_usd	location	size	color	season	review_rating	subscription_status	shipping_type	discount_applied	previous_purchases	payment_method	frequency_of_purchases	age_group	purchase_frequency_days
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	14	Venmo	Fortnightly	Middle-aged	14
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	2	Cash	Fortnightly	Young Adult	14
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	23	Credit Card	Weekly	Middle-aged	7
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	49	PayPal	Weekly	Young Adult	7
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	31	PayPal	Annually	Middle-aged	365

Database Integration

Loaded cleaned DataFrame into **MySQL** for SQL-based analysis.

	customer_id	age	gender	item_purchased	category	purchase_amount_usd	location	size	color	season	review_rating	subscription_status	shipping_type	discount_applied	previous_purchases	payment_method
►	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	14	Venmo
	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	2	Cash
	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	23	Credit Card
	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	49	PayPal

payment_method	frequency_of_purchases	age_group	purchase_frequency_days
Venmo	Fortnightly	Middle-aged	14
Cash	Cash nightly	Young Adult	14
Credit Card	Weekly	Middle-aged	7
PayPal	Weekly	Young Adult	7

Top Products & Shipping Insights

Top 5 Products by Rating

Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

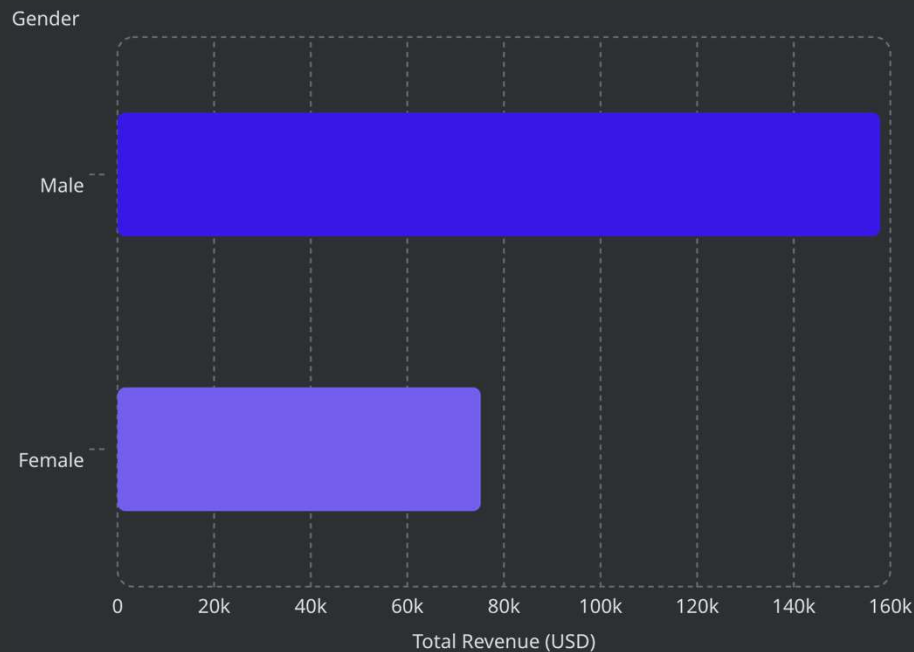
Shipping Type Comparison

Express shipping customers spend slightly more on average (\$60.48) vs. Standard (\$58.46).

Discount-Dependent Products

Hat (50% discounted purchases), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), and Pants (47.37%) rely most heavily on discounts.

Revenue & High-Spending Customers



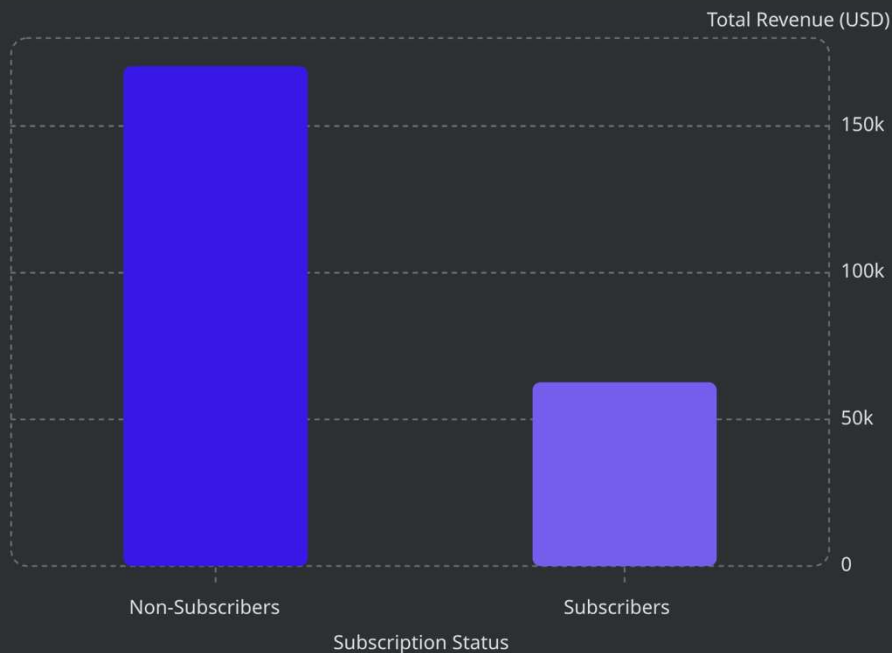
Revenue by Gender

Male customers generated **\$157,890** — over twice the revenue of female customers (**\$75,191**).

High-Spending Discount Users

839 customers used discounts yet spent **above the average purchase amount** — indicating discount-driven loyalty rather than price sensitivity alone.

Subscribers vs. Non-Subscribers



Key Findings

- 73% of customers are non-subscribers (2,847 vs. 1,053)
- Non-subscribers generated **\$170,436** in total revenue vs. **\$62,645** for subscribers
- Average spend is nearly identical: **\$59.87** (non-subscribers) vs. **\$59.49** (subscribers)

i Subscribers show strong loyalty potential — **958 repeat buyers** (5+ purchases) are subscribed.

Customer Segmentation

3,116

Loyal Customers

80% of the base — highest purchase frequency

701

Returning

18% — engaged but not yet fully loyal

83

New Customers

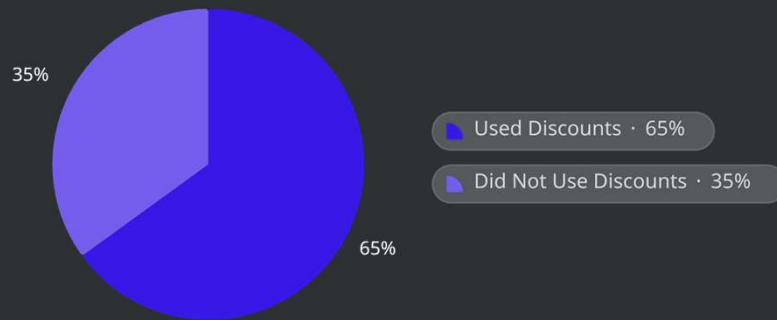
2% — growth opportunity for conversion

Segmentation based on purchase history reveals a **strongly loyal customer base**, with significant room to convert returning and new customers through targeted programs.

Discount Impact Analysis

Analyzing how discounts influence purchasing decisions and customer value is crucial for optimizing promotional strategies.

Discount Usage Rate



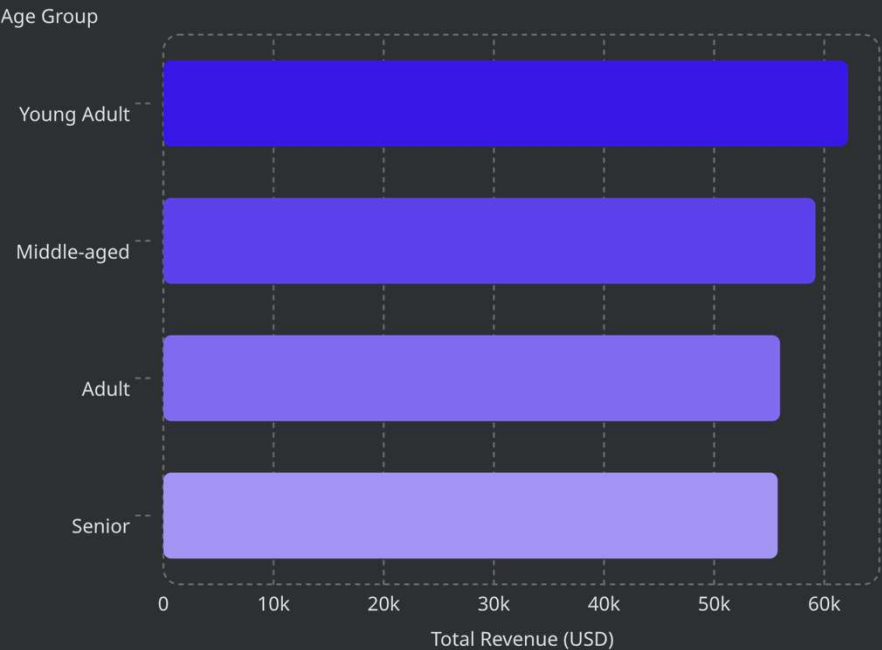
65% of customers engaged with discounts, indicating their significant role in driving customer interaction and purchasing behavior.

Average Spend Comparison



Customers using discounts exhibit a slightly **higher average spend of \$62.10** compared to non-discount users (\$59.00). This suggests discounts may incentivize larger purchases or attract customers with higher spending potential.

Revenue by Age Group

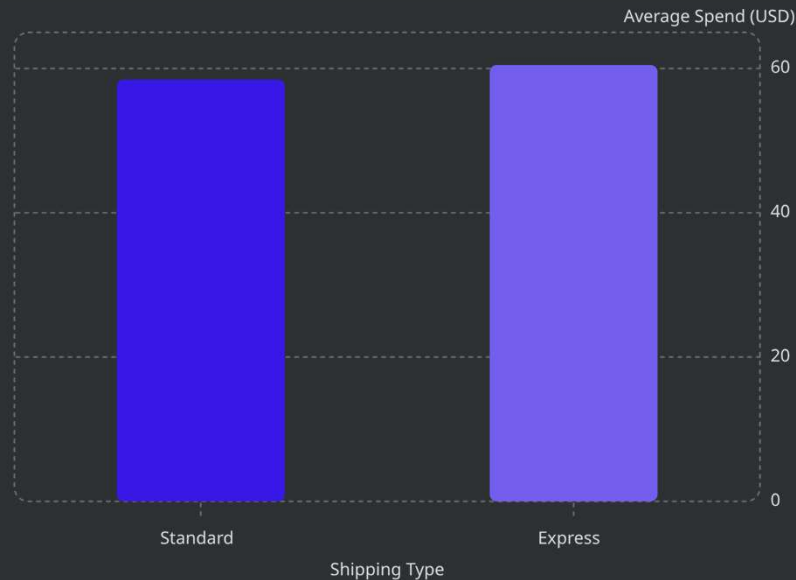


Top 3 Products per Category

Category	#1	#2
Accessories	Jewelry	Sunglasses
Clothing	Blouse	Pants
Footwear	Sandals	Shoes
Outerwear	Jacket	Coat

Young Adults lead revenue at **\$62,143**, though all age groups contribute relatively evenly — a broad, diversified customer base.

Shipping Preference Impact



Customers opting for Express shipping show a slightly higher average transaction value, indicating a propensity to spend more when speed is a priority.

Key Metrics

- 70% of customers utilize Standard shipping.
- 30% of customers prefer Express shipping.
- Express shipping correlates with higher customer satisfaction and increased likelihood of repeat purchases due to faster delivery.

Optimizing Shipping Strategy

01

Promote Express Benefits

Highlight the value of faster delivery and convenience to drive adoption among high-value segments.

02

Tiered Loyalty Programs

Offer free or discounted express shipping as a reward for loyal customers to enhance retention.

03

Dynamic Shipping Options

Implement smart logic that suggests express shipping based on order value or product type, maximizing conversions.

Power BI Dashboard

Key Metrics

3.9K total customers · Avg purchase \$59.76 · Avg review rating 3.8

Subscription Split

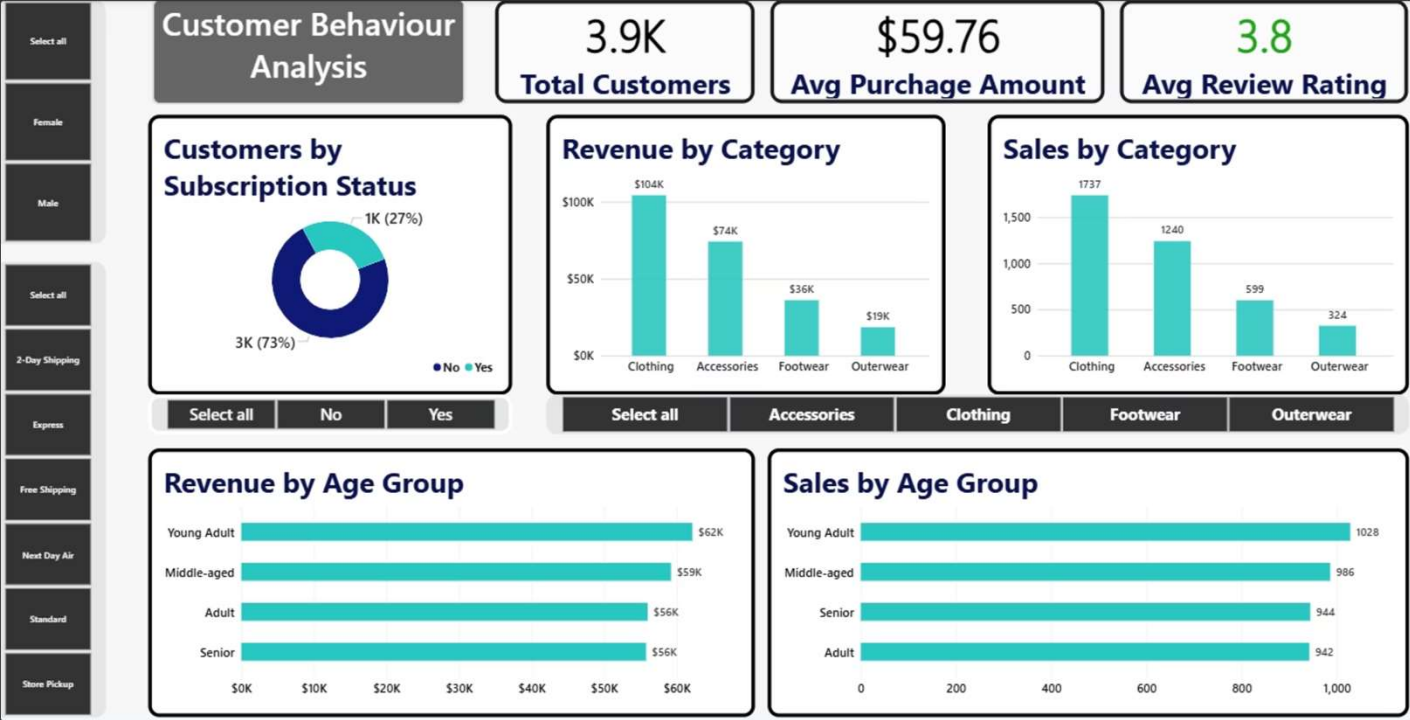
73% non-subscribers (3K) vs. 27% subscribers (1K)

Revenue by Category

Clothing \$104K · Accessories \$74K · Footwear \$36K · Outerwear \$19K

Sales by Age Group

Young Adult 1,028 · Middle-aged 946 · Senior 944 · Adult 942



Business Recommendations

→ Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base.

→ Customer Loyalty Programs

Reward repeat buyers to accelerate movement into the "Loyal" segment.

→ Review Discount Policy

Balance sales boosts with margin control — 5 products rely on ~50% discount rates.

→ Product Positioning & Targeted Marketing

Highlight top-rated products in campaigns; focus on high-revenue age groups and express-shipping users.